
Integrable Systems — Exercise Sheet 12

Scattering without diffraction

Variant: Questions only

Conventions, as in Lecture 12: the open Toda chain is $H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^{N-1} e^{q_n - q_{n+1}}$, with $\{f, g\} = \sum_n (\partial_{q_n} f \partial_{p_n} g - \partial_{p_n} f \partial_{q_n} g)$; its Lax matrix L is symmetric tridiagonal with diagonal p_n and off-diagonal $a_n = e^{(q_n - q_{n+1})/2}$ (Lecture 8), and $x_n = a_n^2 = e^{q_n - q_{n+1}}$. Moser's theorem (cited in Lecture 12): as $t \rightarrow \pm\infty$ the particles separate and move on straight lines $q_n = p_n^\pm t + \beta_n^\pm + o(1)$, with $p_n^+ = \lambda_n$ and $p_n^- = \lambda_{N+1-n}$ for the eigenvalues $\lambda_1 < \dots < \lambda_N$ of L , and the line carrying λ_k is displaced by $D_k = \beta_k^+ - \beta_{N+1-k}^- = \sum_{j>k} 2 \ln(\lambda_j - \lambda_k) - \sum_{j<k} 2 \ln(\lambda_k - \lambda_j)$.

Exercise 1 (the Toda spring). (i) Expand $V(r) = e^{-r} + r - 1$ to fourth order in r , and show that to cubic order it is the FPUT spring $\frac{1}{2}r^2 + \frac{\alpha}{3}r^3$ with $\alpha = -\frac{1}{2}$. (ii) Describe the force $-V'(r)$ for large compression ($r \rightarrow -\infty$) and large stretching ($r \rightarrow +\infty$). (iii) Why can the linear term r be dropped on a ring of springs, and what does dropping it do on the open chain?

Exercise 2 (two particles in closed form). Take $N = 2$, $\bar{p} = \frac{1}{2}(p_1 + p_2)$ and $x = q_1 - q_2$, so that $\ddot{x} = -2e^x$ and $\frac{1}{4}\dot{x}^2 + e^x = \frac{1}{4}v^2$ with $v > 0$ (Lecture 12). (i) Verify that $x(t) = \ln \frac{v^2}{4} - 2 \ln \cosh \frac{v(t-t_0)}{2}$ solves both equations. (ii) Show that the momenta $p_{1,2} = \bar{p} \pm \frac{1}{2}\dot{x}$ are exchanged by the collision. (iii) Show that the eigenvalues of L are $\bar{p} \mp \frac{v}{2}$ at every time. (iv) With $t_0 = 0$ and $q_{1,2} = \bar{p}t \pm \frac{x}{2}$, show that $q_1 \simeq \lambda_2 t + \ln v$ as $t \rightarrow -\infty$ and $q_2 \simeq \lambda_2 t - \ln v$ as $t \rightarrow +\infty$, with $\lambda_2 = \bar{p} + \frac{v}{2}$: the line carrying λ_2 is displaced by $-2 \ln v$. Find the time delay when $\lambda_2 > 0$. (v) What happens to the displacement for $v < 1$?

Exercise 3 (independence, for three particles). For $N = 3$ let $F_k = \text{tr } L^k$, $k = 1, 2, 3$. (i) Compute the Jacobian $\det(\partial F_k / \partial p_n)$, and show that it is $6[(p_2 - p_1)(p_3 - p_1)(p_3 - p_2) + x_1(p_1 - p_2) + x_2(p_2 - p_3)]$. (ii) Deduce that dF_1, dF_2, dF_3 are independent on a dense open set.

Exercise 4 (the first charges). For general N : (i) Show that $\text{tr } L = \sum_n p_n$, $\text{tr } L^2 = 2H$ and $\text{tr } L^3 = \sum_n p_n^3 + 3 \sum_{n=1}^{N-1} x_n(p_n + p_{n+1})$. (ii) Find the flows of $\text{tr } L$ and of $\text{tr } L^3$: how does each move the positions? (iii) Why did nobody see $\text{tr } L^3$ as the generator of a symmetry of the positions?

Exercise 5 (asymptotic momenta, and three bodies). (i) Assuming Moser's separation, $q_{n+1} - q_n \rightarrow +\infty$ as $t \rightarrow \pm\infty$, show that the asymptotic momenta are the eigenvalues of L , in increasing order along the chain as $t \rightarrow +\infty$ and decreasing as $t \rightarrow -\infty$. (ii) For three particles with $\lambda = (-3, 0, 3)$, compute D_1, D_2, D_3 , and write each as a sum of two-body displacements of Exercise 2, one per pair. (iii) Show that $\sum_k D_k = 0$ for every spectrum. What does this say about the centre of mass?

Exercise 6 (five particles, computed). (Computational.) Take $N = 5$, $q(0) = (-1, -\frac{1}{2}, 0, \frac{1}{2}, 1)$ and $p(0) = (2, 1, 0, -1, -2)$. (i) Integrate the equations of motion from $t = -40$ to $t = 40$, and check that $\text{tr } L^k$, $k = 1, \dots, 5$, stay constant, and that $H = \frac{1}{2} \sum_k \lambda_k^2$. (ii) Compare $p(\pm 40)$ with the eigenvalues of L . (iii) Fit the asymptotic lines, $\beta_n^\pm = q_n(\pm 40) - p_n(\pm 40) \cdot (\pm 40)$, and compare the displacements D_k with Moser's formula.

Exercise 7 ★. *For two particles the fibre is a plane.* Lecture 12's reading proves, in outline, that every connected component of a regular level set of the open chain is a copy of $\mathbb{R}^N$. Do it for $N = 2$. Let $\Sigma = \{p_1 + p_2 = 2\bar{p}, H = \bar{p}^2 + \frac{1}{4}\nu^2\}$ with $\nu > 0$, and use $Q = \frac{1}{2}(q_1 + q_2)$, $x = q_1 - q_2$ and $p_x = \frac{1}{2}(p_1 - p_2)$. (i) Show that (Q, p_x) , with $|p_x| < \nu/2$, are coordinates on Σ , so that $\Sigma \cong \mathbb{R}^2$. (ii) Show that the flow of $p_1 + p_2$ moves Q at unit speed and fixes x and p_x , and that along the flow of H , p_x strictly decreases. (iii) Deduce that no combination $s(p_1 + p_2) + tH$ with $(s, t) \neq (0, 0)$ returns a point of Σ to itself: the stabiliser of Lecture 5's $\mathbb{R}^2$ -action is trivial.